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Predicting the Salary of an NHL Player

Abstract

This project aims to find the best predictive method to predict the salary of an NHL player. The escalating salaries in professional sports leagues, including the NHL, have made accurate salary prediction a critical endeavor for team management, agents, and players alike. Using a dataset consisting of player statistics and performance metrics, this study will investigate the efficacy of several prediction models. To decide which is the best, each method's ability will be judged using cross-validation to explore which is the best to predict the salary of an NHL player by looking which model has the lowest estimated test mean squared error. The project will help give an idea of how useful predictive analysis can be in professional sports, making it easier for all parties involved to make well-informed decisions.

Introduction

In sports, whenever one wishes to determine the level of a player, they always look at their salary. Salary is that one measurement which the higher it is indicates how big of an asset the player is to the team. Reports are made if the player gets signed with a record-breaking value in several hundred millions of dollars. In the current age of big data, records of every player

statistic are created and analyzed, making them increasingly transparent. Hockey is one such sport that generates numerous different statistics creating more opportunity for analysis in the National Hockey League (NHL). However, NHL teams have a salary cap and must hire a certain number of players within that budget. This cap forces teams to carefully manage their money to assemble the best possible team, which attracts fans and increases revenue. Much like any other firm, it is in the best interest of NHL team management to allocate their limited resources efficiently to build the most effective roster of players possible. However, the players also want to receive the financial compensation they deserve for their high level of skills and services, especially since a career in the NHL does not last long. (Tonack) Hence, the goal of this project is to devise the best possible way to predict a salary of an NHL player. By using some proper prediction methods, teams can prevent overspending on player contracts and thus build a more competitive team. In this project, several regression methods such as linear regression, decision trees, random forests, ridge, and lasso regression, etc. are used for a set of predictors to predict the variable salary. Then, cross-validation was conducted for each type of regression model to judge how accurate the model was in predicting out of sample data. Austin Tonack's research on predicting salaries of NHL players based on player performance also brings an interesting point. He mentions his motivation was the Stanley Cup in 2017-18, which helped increase the valuation of the Washington Capitals, a team franchise in the NHL, from \$100 million to \$625 million only because they effectively chose the best possible players due to player's best statistic being their success in scoring goals and creating opportunities while they're on ice. Tonack also creates a regression model to predict the salary of an NHL player based on their performance compared to the player's actual salary to see which players are over or under paid. His research brings great motivation for this project as the results of that analysis indicated that usually rookies who

contributed a great deal to the team were underpaid by almost two million dollars. This empirical project thus is also related as its goal is to create the best possible model to predict an NHL player's salary to ensure fair compensation and benefit teams to increase their valuation as they can effectively select their players. In the research project done by Northwestern University, Louis-Charles G  n  reux and Allen Xu also sought to create a model that efficiently predicts player salary. Something interesting they mentioned is historically in most predictions of player salaries in NHL, one of the most common metrics used in the field of hockey player evaluation has been plus-minus (the goal difference between goals for and against while a player is on the ice. However, they point out that although this metric is simple to compute, it is largely dependent on the performance and strength of the whole team instead of individual players, often leading to limitations in predictions. Keeping this in mind, this empirical project focuses on using less short-term predictors for the regression models and more player-based statistics which can lead to a more accurate prediction of the salary. Ryan McGinnis, who also did research to determine the value of an NHL player found in his results that a lot of rookies were being underpaid, and the team's players he focused on, New York Islanders, are underperforming and the franchise is losing out on money that could be used in another part of their organization. The motivation behind this project seems clearer on the need to creating an accurate model to be able to predict the salary of players to ensure teams are spending their budgets in the most efficient way possible. Gu et.al's paper on devising a prediction model for NHL player salary seemed to be more different. They used a blend of team-based and individual player-based predictors and devised a model that has a 94.05% precision in predicting salary. This does contradict previous studies but is related to the empirical project as a high precision rate indicates that when choosing predictors, it may be important to consider both team-based and individual player-

based factors to improve the accuracy of the salary prediction model. Fullard's study also concurs with this as his study on predicting the salary of NHL goalies specifically, reveals individual goalie performance measures do not relate to salary, but do relate to overall team records. This proves that hockey has characteristics of reciprocal interdependence. Reciprocal interdependent sports refer to team sports in which a number of players' outputs are needed for an outcome (Fullard). His study concluded that hierarchical pay structures better predict salaries, going against some of the previous few studies, meaning that salary given based on things like performance, seniority, experience seem to be more accurate in predicting a player's salary. This idea stays relevant for the empirical project as understanding the intricate factors influencing salary allocation not only sheds light on the dynamics of professional sports economics but also has practical implications for league policies.

Data description

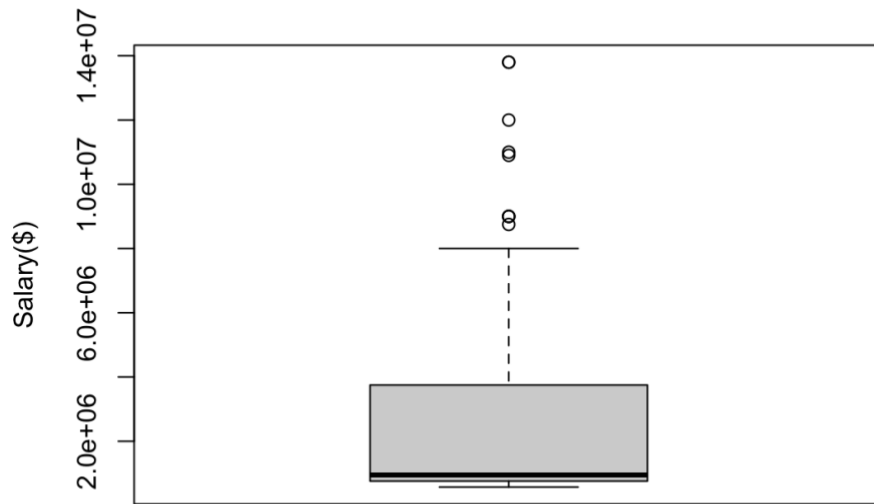
The dataset features the salaries of 874 NHL players from the 2016-17 season. The data was obtained from Kaggle, a popular open data website. The original author who compiled the data gathered the raw data from a website called Hockey Abstract. Hockey Abstract is a collection of books that collects raw data for different player statistics and performance metrics in the NHL. The initial dataset contains 874 observations, 153 inputs, and 1 output. However, since a lot of rows were blanks in the dataset, I removed those observations to make the data cleaner making it easier to use and conduct prediction analysis with. After cleaning the data using the R software, the new dataset, which was used for prediction analysis, has 364 observations with the same 153 inputs, and 1 output. The dataset includes several different variables for predicting salary; however, for the purpose of this project, the predictors chosen were the following:

Variable	Description
GP	Games Played
CF	The team's shot attempts while this player was on the ice
A1	First assists, primary assists
A2	Second assists, secondary assists
PTS	Points. Goals plus all assists
TOI	Time on ice, in minutes, or in seconds
CA	Shot attempts allowed while this player was on the ice
DSA	Dangerous shots allowed while this player was on the ice
DPS	Defensive point shares
G	Goals

Data

The inputs of this data are a mixture of categorical data (country born, year drafted, etc.), discrete data (goals scored, penalties incurred, faceoffs won, etc.) and continuous data. To see an overall trend of the salaries in the NHL season of 2016-17, we can create a box plot:

Figure 1: Boxplot of Salaries of NHL players in the 2016-17 season



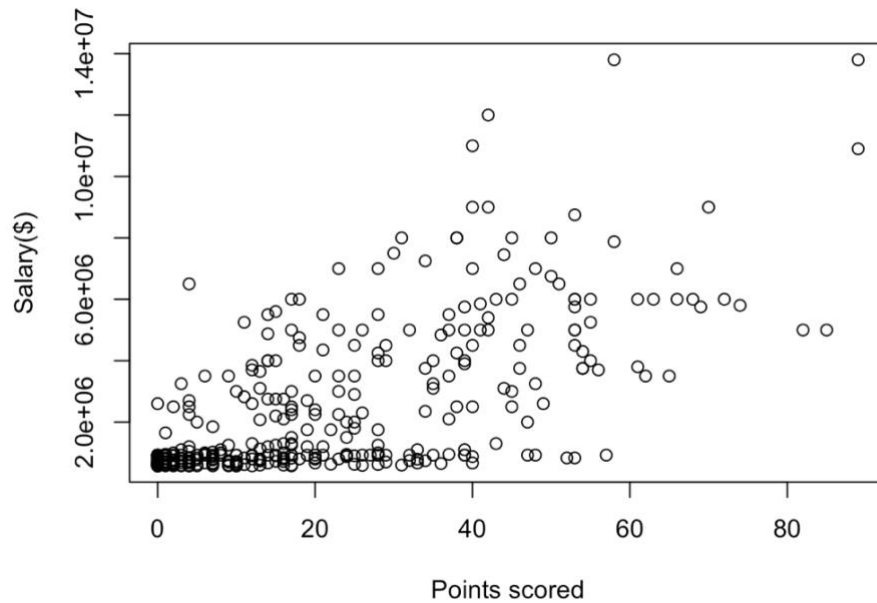
The minimum salary of the players in the 2016-2017 NHL season is \$575,000, the average salary is \$2,325,289, and the largest salary is \$14,000,000, while the median salary is \$950,000.

As seen, with the minimum salary being \$575,000. This indicates that during the season no team is allowed to pay any player less than this amount, which is decided on at the start of the season. The salary cap for each team during the given season was \$73,000,000. This means that this amount must cover all the players' salaries on the team. The maximum salary that a player may receive is 20% of the team's salary cap as per the NHL rules. Using this information, we can determine the range of our output variable, Salary. The minimum a player can earn in the given season is \$575,000 and the maximum a player can earn is \$14,600,000, which is 20% of the salary cap. However, no player for the season received the full amount as their salary because the highest value was \$14,000,000. This rightward skewness in the salary indicates that a limitation that our model may have from the dataset is that it will be more accurate in predicting the salaries of lower salary players compared to higher salary ones. The predictors chosen were the

ones which seemed to be more correlated with the output variable of salary. For instance, the predictor “points” (PTS) had a correlation coefficient of 0.672, indicating a positive correlation.

We can also see this in a scatter plot form:

Figure 2: Scatter plot of relation between PTS predictor and Salary



As seen in the plot above, there seems to be a generally positive correlation as the more points scored, the higher the salary of the player. This also makes sense because in the NHL the more points a player scores, the more their value increases as the more they are contributing to the team. This causes the franchise to spend more on the player to retain them or acquire them to benefit their squad. For this project a regression model was used with the chosen predictors following the base linear equation model: $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \dots + \beta_p X_p + \epsilon$. The goal for our project is of course to predict Y , which will be $\hat{Y}$.

Results

To answer our question and implement different prediction methods, we first can implement the simplest method to predict, which is linear regression. To conduct cross validation using the split-half approach, we divide the data into a test data and training data. Training data is our in-sample data, and we use this data to fit an estimate of the linear model. Once we create the model, we can then use the test data to create an estimate of the Mean Squared Error (MSE), which is the difference between our actual value and the predicted value. We can then find the mean of this for all our values and find the estimated Test MSE. Therefore, our linear model that we fit for our training set for $\widehat{Salary}$, the outcome variable is:

$$\begin{aligned}\widehat{Salary} = & \widehat{\beta}_0 + \widehat{GP}\widehat{X} + \widehat{CF}\widehat{X}_2 + \widehat{A1}\widehat{X}_3 + \widehat{A2}\widehat{X}_4 + \widehat{PTS}\widehat{X}_5 + \widehat{TOI}\widehat{X}_6 + \widehat{CA}\widehat{X}_7 + \widehat{DSA}\widehat{X}_8 \\ & + \widehat{DPS}\widehat{X}_9 + \widehat{G}\widehat{X}_{10}\end{aligned}$$

By fitting our model for Salary using the training data and then using the test data, we conduct cross validation and finding the average for each observation, we get a value of 2.8188×10^{12} for our estimated Test MSE value. We can also do OLS regression by including a polynomial, we can also get a new equation as our PTS variable had a high correlation with salary, we can also use $\widehat{PTS}^3$. Using this polynomial we can conduct another split-half cross-validation approach in the end giving us an estimated Test MSE value of 2.8029×10^{12} , which is slightly better than just regular OLS regression.

We can also conduct our OLS regression with Step functions. In Step functions we create new variables by creating certain amounts of cut points for our variable to split the variable further into bins and fit a different constant for each bin. In this regression's case, we choose to split the variable "PTS", where we create four cut points: C1, C2, C3, C4. With these cut points, we create four new variables for the values in the variable PTS. This means that if the value of

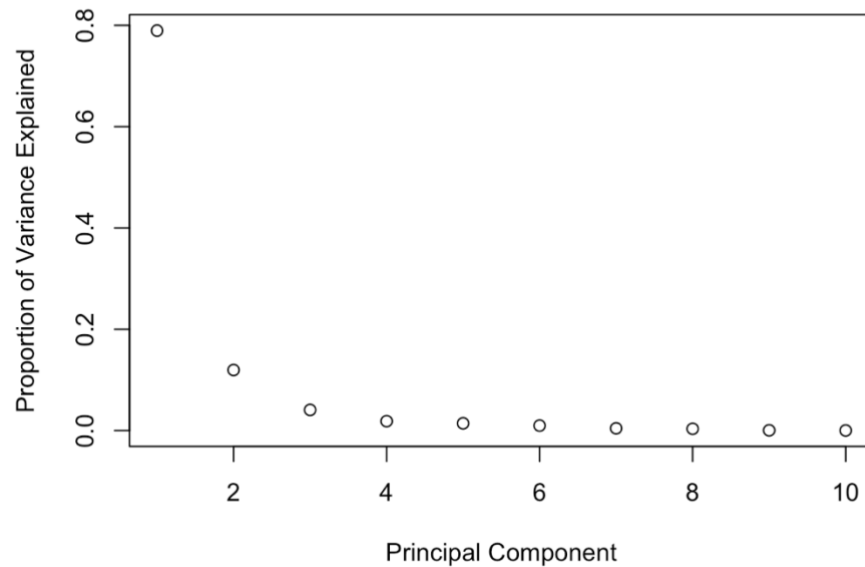
PTS will take a value that is less than C1, our first cut point we get a different value than if it is between C2 and C1. Hence, after cutting the PTS variable we get the following: 22.2 as our C1, 44.5 as C2, 66.8 as C3, and 89.1 as C4. Using these cut points, we can again conduct cross validation to see if these splits help us better predict player salaries. Upon conducting cross-validation, we get an estimated Test MSE of 2.644×10^{12} . Out of all the OLS techniques we tried, the best method so far is through using step functions as it provides us with the lowest estimated Test MSE value compared to using polynomials or just a simple linear regression model.

Another useful prediction method is Lasso regression. Lasso adds a penalty for the complexity of the model to the absolute value of the magnitude of the coefficients. This penalty term often has a strong effect as in this regression, it automatically performs variable selection, that is in Lasso certain predictors will be exactly equal to zero when the tuning parameter, lambda, is sufficiently large. Hence, it only uses a subset of variables. To choose the best tuning parameter for our model, we once again conduct cross-validation using the split-half approach. We want to select the value of lambda which minimizes the estimated Test MSE. We use cross-validation on a generalized model for our lambda and save that value as our best lambda. For our model, we get the best lambda value of 25267.3. Using this value, we simply conduct Lasso regression, where we estimate a model based on this lambda, and use cross-validation to calculate the estimated Test MSE, our $\hat{Y}_0$, for which the value we got is 3.1748×10^{12} .

We can also conduct principal component analysis to our model. This is where we transform a model matrix of all our variables into a smaller one, without losing much information. We do this by conducting linear combinations of existing variable X to create new variable Z for m

variables. We thus can see the total proportion of variance explained when conducting principal component analysis as we see the ratio between the sample variance for our new variables Z and the total variance from all our variables $X_1, X_2, X_3, X_4, \dots, X_p$. Applying this idea to our dataset, we get the following plot:

Figure 3: Proportion of Variance explained

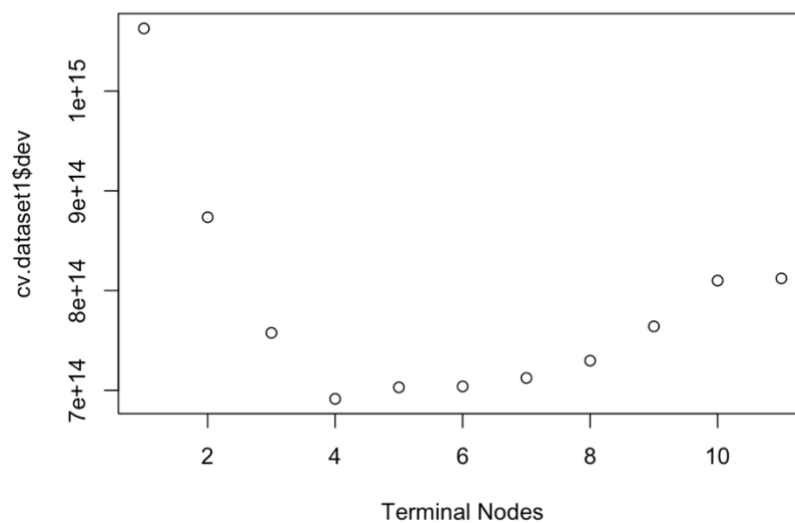


As we can see in the graph, using one principal component explains almost 0.8 of variance while the higher we go the lower the explained proportion of variance is.

Another method to predict is using Regression trees. These are usually easy to interpret and see visually; however, they often are poor in the accuracy of the prediction. The way regression tree works in general is for a set of variables, we choose a variable to split our observations. For instance, if for a set of 5 variables, X_1, X_2, X_3, X_4 , and X_5 , we choose X_3 to be the variable that will split our observations, we can then split our entire observations based on if the value of the other variables are either greater or smaller than a certain value of X_3 . We can

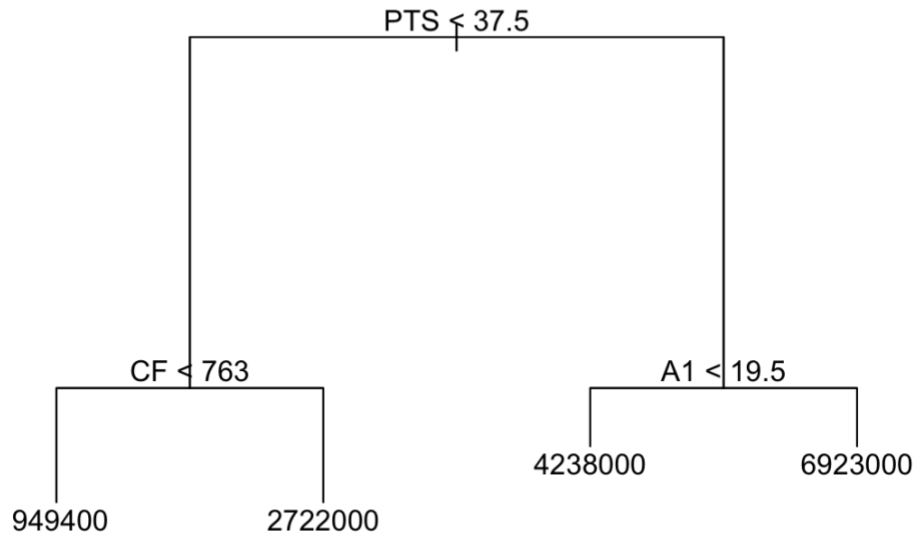
split our observations however many times as we desire, and for each region we can compute its sample mean of the response variable. To choose the best number of splits, we can do tree pruning. This is done by training a tree on the training dataset by splitting regions where for a set of variables, we choose to minimize our training Residual Sum of Squares (RSS) among our set of variables and decide the cut point. However, the model we minimize over also has a penalty term like Lasso which is alpha for the number of terminal nodes we have. To choose alpha, we again conduct cross-validation. In the case of NHL salaries, to find the best amount of splits we can conduct cross-validation upon which we see:

Figure 4: Cross-validation estimates on best Terminal Node value



As we can see from the data, when the value of the node becomes too large, the estimated Test MSE becomes higher; hence, the best value we can see is a value of 4, so 4 splits. From this we can also conduct a regression tree for our data, which would have 4 splits, looking something like this:

Figure 5: Regression Tree with 4 Terminal Nodes



As we can see, it makes it extremely easy to interpret and predict the salary value for our observations. For instance, if a player were to score more than 37.5 points, then we can further split the data by their primary assists (A1), and if that is more than 19.5, then their predicted salary will be \$6,923,000; however, if the A1 is smaller than 19.5, then their salary would be \$4,238,000. However, by making a very over simple model, it means that our error rate is bound to be very high, and a lot of predicted salary values will be inaccurate. This is why the estimated Test MSE value we get using Regression Trees and Pruning is 3.9121×10^{12} . This value is quite a lot higher than when we did Lasso or even when we did different OLS regression techniques.

A good way to reduce this inaccuracy is by conducting random forests. Random forests create a certain number of bootstrapped datasets, and we randomly select the predictors among

all our predictors in the dataset and then minimize the training RSS by selecting a cut point among our set of predictors similar to Regression Trees. By randomly selecting predictors, we also reduce the risk of overfitting as when we are minimizing training RSS, there is always a risk of overfitting. This way, we force different trees to use different predictors, so the resulting trees aren't too similar. Applying this to our dataset, we can conduct Random Forest by using the relevant library in R-studio and for 6 predictors as we are only using a subset of predictors. This gives us an estimated Test MSE value of 3.4699×10^{12} , slightly better than Regression Trees, but still not as good as Lasso or OLS.

Conclusion

From our analysis, we can decipher that based on the estimated Test MSE value of 2.644×10^{12} that the best method to predict is using step functions. While the other methods weren't as good, the values of the estimated Test MSE when conducting Lasso and polynomial regression, were close to the ones we got with step functions. With Lasso our value of the estimated Test MSE was 3.1748×10^{12} while with polynomial regression our value was 2.8029×10^{12} . Overall, OLS regression did turn out to be a bit better for this dataset model. This also makes sense because a lot of the predictors in this dataset did seem to have a strongly positive or negative correlation with Salary. Using the Step functions model, we can also predict for a set of values given for our predictors, where we get:

$$\widehat{Salary} = \widehat{\beta}_0 + \widehat{GP}\widehat{X} + \widehat{CF}\widehat{X}_2 + \widehat{A1}\widehat{X}_3 + \widehat{A2}\widehat{X}_4 + \widehat{PTS}_1\widehat{X}_5 + \widehat{PTS}_2\widehat{X}_6 + \widehat{PTS}_3\widehat{X}_6 + \widehat{TOI}\widehat{X}_7 \\ + \widehat{CA}\widehat{X}_8 + \widehat{DSA}\widehat{X}_9 + \widehat{DPS}\widehat{X}_{10} + \widehat{G}\widehat{X}_{11}$$

Where PTS1, PTS2, and PTS3 are dummy variables. By putting in our coefficients and certain values for our X in each one, we get:

$$\begin{aligned}\widehat{Salary} = & 755186.95 - 54164.87(81) + 1962.60(500) + 65800.61(10) - 20486.87(13) \\ & + 0 + 788370.75(70) + 0 + 41.13(63592) + 1003.19(800) \\ & - 3649.83(150) - 491685.21(1.5) + 56292.67(20)\end{aligned}$$

55,384,955.005. This doesn't make proper sense as the maximum value we were able to get for this season was around \$14,600,000. However, what explains this is that I did choose high amounts for certain variables as it is uncommon for a player to be able to do all that in one season unless they are extremely talented. Hence, the salary value is quite high.

References

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Appendix

Summary Statistics Output:

Salary	Born	City
Length:364	Length:364	Length:364
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

Pr/St	Cntry	Nat
Length:364	Length:364	Length:364
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

Ht	Wt	DftYr	DftRd
Min. :67.00	Min. :160	Min. :1997	Min. :1.000
1st Qu.:72.00	1st Qu.:191	1st Qu.:2006	1st Qu.:1.000
Median :73.00	Median :202	Median :2009	Median :2.000
Mean :73.06	Mean :202	Mean :2009	Mean :2.805
3rd Qu.:75.00	3rd Qu.:212	3rd Qu.:2012	3rd Qu.:4.000
Max. :78.00	Max. :244	Max. :2016	Max. :9.000

Ovrl	Hand	Last Name
Length:364	Length:364	Length:364
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

First Name	Position	Team
Length:364	Length:364	Length:364
Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character

GP	G	A	A1
Min. : 1.00	Min. : 0.000	Min. : 0.00	Min. : 0.000
1st Qu.:24.00	1st Qu.: 1.000	1st Qu.: 2.00	1st Qu.: 1.000
Median :65.00	Median : 5.000	Median :10.00	Median : 5.000
Mean :52.72	Mean : 7.838	Mean :13.17	Mean : 7.264
3rd Qu.:79.00	3rd Qu.:12.000	3rd Qu.:21.00	3rd Qu.:11.000
Max. :82.00	Max. :44.000	Max. :55.00	Max. :36.000

A2	PTS	+/-	E+/-
Min. : 0.000	Min. : 0.00	Min. : -31.0000	Min. : -19.00000

1st Qu.: 1.000 1st Qu.: 4.00 1st Qu.: -6.0000 1st Qu.: -3.02500
 Median : 4.000 Median :16.00 Median : -1.0000 Median : -0.40000
 Mean : 5.909 Mean :21.01 Mean : -0.4423 Mean : -0.06923
 3rd Qu.: 9.000 3rd Qu.:34.25 3rd Qu.: 5.0000 3rd Qu.: 2.40000
 Max. :28.000 Max. :89.00 Max. : 34.0000 Max. : 20.30000

PIM Shifts TOI TOIX

Min. : 0.0 Min. : 13.0 Min. : 429 Min. : 7.2
 1st Qu.: 10.0 1st Qu.: 438.2 1st Qu.: 18885 1st Qu.: 314.8
 Median : 24.0 Median :1288.5 Median : 56860 Median : 948.7
 Mean : 28.2 Mean :1155.5 Mean : 52785 Mean : 875.6
 3rd Qu.: 38.0 3rd Qu.:1797.0 3rd Qu.: 83795 3rd Qu.:1377.7
 Max. :154.0 Max. :2657.0 Max. :133550 Max. :2218.9

TOI/GP TOI/GP_1 TOI% IPP%

Min. : 6.75 Min. : 6.75 Min. :13.10 Min. : 0.00
 1st Qu.:12.17 1st Qu.:12.15 1st Qu.:22.88 1st Qu.: 34.10
 Median :15.37 Median :15.36 Median :27.30 Median : 54.30
 Mean :15.37 Mean :15.35 Mean :27.51 Mean : 49.78
 3rd Qu.:18.39 3rd Qu.:18.38 3rd Qu.:32.30 3rd Qu.: 67.65
 Max. :27.15 Max. :27.12 Max. :44.90 Max. :100.00

SH% SV% PDO F/60

Min. : 0.000 Min. :0.667 Min. : 750.0 Min. : 0.000
 1st Qu.: 6.300 1st Qu.:0.904 1st Qu.: 978.0 1st Qu.: 1.677
 Median : 8.000 Median :0.916 Median : 997.5 Median : 2.270

Mean : 7.805 Mean : 0.915 Mean : 993.0 Mean : 2.269

3rd Qu.: 9.600 3rd Qu.: 0.927 3rd Qu.: 1016.0 3rd Qu.: 2.982

Max. : 40.000 Max. : 1.000 Max. : 1257.0 Max. : 10.780

A/60 Pct% Diff Diff/60

Min. : 0.000 Min. : 0.00 Min. : -44.000 Length:364

1st Qu.: 2.070 1st Qu.: 39.05 1st Qu.: -7.000 Class :character

Median : 2.470 Median : 48.60 Median : -1.000 Mode :character

Mean : 2.545 Mean : 45.93 Mean : 1.742

3rd Qu.: 2.870 3rd Qu.: 56.73 3rd Qu.: 9.250

Max. : 16.740 Max. : 100.00 Max. : 61.000

iCF iCF_2 iFF

Length:364 Length:364 Length:364

Class :character Class :character Class :character

Mode :character Mode :character Mode :character

iSF iSF_3 iSF_4 ixG

Length:364 Min. : 0.00 Min. : 0.00 Min. : 0.000

Class :character 1st Qu.: 24.75 1st Qu.: 24.75 1st Qu.: 1.800

Mode :character Median : 81.00 Median : 81.00 Median : 5.800

Mean : 88.88 Mean : 88.90 Mean : 7.914

3rd Qu.: 136.25 3rd Qu.: 136.25 3rd Qu.: 12.100

Max. :302.00 Max. :302.00 Max. :33.000

iSCF iRB iRS iDS

Length:364 Length:364 Min. : 0.000 Min. : 0.00

Class :character Class :character 1st Qu.: 2.000 1st Qu.: 3.00

Mode :character Mode :character Median : 6.000 Median :11.00

Mean : 7.629 Mean :13.94

3rd Qu.:12.000 3rd Qu.:20.25

Max. :32.000 Max. :63.00

sDist sDist_5 Pass iHF

Length:364 Length:364 Min. : 0.00 Min. : 0.00

Class :character Class :character 1st Qu.: 34.58 1st Qu.: 24.00

Mode :character Mode :character Median :117.65 Median : 57.50

Mean :140.38 Mean : 68.48

3rd Qu.:222.47 3rd Qu.: 94.00

Max. :501.20 Max. :364.00

iHF_6 iHA iHDf

Length:364 Length:364 Min. : -114.000

Class :character Class :character 1st Qu.: -17.000

Mode :character Mode :character Median : 1.000

Mean : 6.187

3rd Qu.: 22.000

Max. : 227.000

iMiss iGVA iTKA iBLK

Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.00
 1st Qu.: 11.00 1st Qu.: 6.00 1st Qu.: 4.00 1st Qu.: 11.00
 Median : 32.00 Median : 21.00 Median : 16.00 Median : 29.00
 Mean : 34.32 Mean : 24.65 Mean : 19.29 Mean : 43.74
 3rd Qu.: 54.00 3rd Qu.: 39.00 3rd Qu.: 30.00 3rd Qu.: 61.00
 Max. : 109.00 Max. : 102.00 Max. : 96.00 Max. : 213.00

iGVA_7 iTKA_8 iBLK_9 BLK%

Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.000
 1st Qu.: 6.00 1st Qu.: 4.00 1st Qu.: 11.00 1st Qu.: 2.800
 Median : 21.00 Median : 16.00 Median : 29.00 Median : 4.400
 Mean : 24.57 Mean : 19.24 Mean : 43.65 Mean : 5.083
 3rd Qu.: 39.00 3rd Qu.: 30.00 3rd Qu.: 61.00 3rd Qu.: 7.125
 Max. : 102.00 Max. : 96.00 Max. : 213.00 Max. : 16.700

iFOW iFOL iFOW_10 iFOL_11

Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.00
 1st Qu.: 0.00 1st Qu.: 0.00 1st Qu.: 0.00 1st Qu.: 0.00
 Median : 2.00 Median : 2.00 Median : 2.00 Median : 2.00
 Mean : 85.47 Mean : 85.12 Mean : 85.29 Mean : 84.92
 3rd Qu.: 39.50 3rd Qu.: 45.25 3rd Qu.: 39.50 3rd Qu.: 45.25
 Max. : 1089.00 Max. : 906.00 Max. : 1083.00 Max. : 906.00

FO% %FOT dzFOW dzFOL

Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.00
 1st Qu.: 0.00 1st Qu.: 0.00 1st Qu.: 0.00 1st Qu.: 0.00

Median : 33.30 Median : 0.85 Median : 0.00 Median : 0.00

Mean : 28.93 Mean : 19.30 Mean : 28.29 Mean : 28.62

3rd Qu.: 50.00 3rd Qu.: 24.85 3rd Qu.: 8.00 3rd Qu.: 10.25

Max. : 100.00 Max. : 99.20 Max. : 429.00 Max. : 344.00

nzFOW nzFOL ozFOW ozFOL

Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.00

1st Qu.: 0.00 1st Qu.: 0.00 1st Qu.: 0.00 1st Qu.: 0.00

Median : 0.00 Median : 0.00 Median : 1.00 Median : 1.00

Mean : 27.20 Mean : 27.38 Mean : 29.95 Mean : 29.09

3rd Qu.: 11.25 3rd Qu.: 12.25 3rd Qu.: 16.25 3rd Qu.: 18.00

Max. : 324.00 Max. : 326.00 Max. : 420.00 Max. : 390.00

FOW.Up FOL.Up FOW.Down FOL.Down

Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.00

1st Qu.: 0.00 1st Qu.: 0.00 1st Qu.: 0.00 1st Qu.: 0.00

Median : 0.00 Median : 0.00 Median : 0.00 Median : 1.00

Mean : 26.04 Mean : 25.25 Mean : 28.47 Mean : 28.93

3rd Qu.: 13.00 3rd Qu.: 14.00 3rd Qu.: 12.25 3rd Qu.: 16.25

Max. : 385.00 Max. : 287.00 Max. : 329.00 Max. : 302.00

FOW.Close FOL.Close OTG 1G

Min. : 0.00 Min. : 0.00 Min. : 0.000 Min. : 0.000

1st Qu.: 0.00 1st Qu.: 0.00 1st Qu.: 0.000 1st Qu.: 0.000

Median : 1.00 Median : 2.00 Median : 0.000 Median : 1.000

Mean : 53.13 Mean : 52.80 Mean : 0.217 Mean : 1.481

3rd Qu.: 25.50 3rd Qu.: 29.25 3rd Qu.:0.000 3rd Qu.: 2.000

Max. :679.00 Max. :549.00 Max. :5.000 Max. :12.000

GWG ENG PSG PSA

Min. :0.000 Min. :0.0000 Min. :0.00000 Min. :0.00000

1st Qu.:0.000 1st Qu.:0.0000 1st Qu.:0.00000 1st Qu.:0.00000

Median :1.000 Median :0.0000 Median :0.00000 Median :0.00000

Mean :1.308 Mean :0.3599 Mean :0.01374 Mean :0.05495

3rd Qu.:2.000 3rd Qu.:0.0000 3rd Qu.:0.00000 3rd Qu.:0.00000

Max. :9.000 Max. :4.0000 Max. :1.00000 Max. :1.00000

G.Bkhd G.Dflct G.Slap G.Snap

Min. : 0.0000 Min. :0.000 Min. : 0.0000 Min. : 0.000

1st Qu.: 0.0000 1st Qu.:0.000 1st Qu.: 0.0000 1st Qu.: 0.000

Median : 0.0000 Median :0.000 Median : 0.0000 Median : 0.000

Mean : 0.7308 Mean :0.228 Mean : 0.9231 Mean : 1.266

3rd Qu.: 1.0000 3rd Qu.:0.000 3rd Qu.: 1.0000 3rd Qu.: 2.000

Max. :10.0000 Max. :3.000 Max. :12.0000 Max. :13.000

G.Tip G.Wrap G.Wrst CBar

Min. :0.0000 Min. :0.00000 Min. : 0.000 Min. :0.0000

1st Qu.:0.0000 1st Qu.:0.00000 1st Qu.: 0.000 1st Qu.:0.0000

Median :0.0000 Median :0.00000 Median : 2.000 Median :0.0000

Mean :0.8379 Mean :0.08516 Mean : 3.755 Mean :0.3242

3rd Qu.:1.0000 3rd Qu.:0.00000 3rd Qu.: 6.000 3rd Qu.:1.0000

Max. :9.0000 Max. :2.00000 Max. :21.000 Max. :6.0000

Post	Over	Wide	S.Bkhd
Min. :0.000	Min. : 0.00	Min. : 0.00	Min. : 0.000
1st Qu.:0.000	1st Qu.: 1.00	1st Qu.: 9.00	1st Qu.: 1.000
Median :1.000	Median : 3.00	Median :27.00	Median : 5.000
Mean :1.437	Mean : 3.39	Mean :29.17	Mean : 7.184
3rd Qu.:2.000	3rd Qu.: 5.00	3rd Qu.:45.00	3rd Qu.:12.000
Max. :8.000	Max. :18.00	Max. :98.00	Max. :44.000

S.Dflet	S.Slap	S.Snap	S.Tip
Min. : 0.000	Min. : 0.00	Min. : 0.00	Min. : 0.000
1st Qu.: 0.000	1st Qu.: 2.00	1st Qu.: 3.00	1st Qu.: 0.000
Median : 0.000	Median : 8.00	Median :10.00	Median : 1.500
Mean : 1.214	Mean :15.96	Mean :14.01	Mean : 4.393
3rd Qu.: 2.000	3rd Qu.: 21.00	3rd Qu.:20.00	3rd Qu.: 7.000
Max. :18.000	Max. :141.00	Max. :77.00	Max. :41.000

S.Wrap	S.Wrst	iPenT	iPenD
Min. :0.0000	Min. : 0.00	Min. : 0.00	Min. : 0.000
1st Qu.:0.0000	1st Qu.: 12.00	1st Qu.: 4.00	1st Qu.: 2.000
Median :0.0000	Median : 40.00	Median :10.00	Median : 7.000
Mean :0.9505	Mean : 45.18	Mean :11.34	Mean : 9.159
3rd Qu.:1.0000	3rd Qu.: 68.25	3rd Qu.:16.00	3rd Qu.:14.000
Max. :9.0000	Max. :182.00	Max. :48.00	Max. :47.000

iPENT2	iPEND3	iPenDf	NPD
Min. : 0.00	Min. : 0.00	Min. :-28.000	Min. :-19.4000

1st Qu.: 4.00 1st Qu.: 2.00 1st Qu.: -6.000 1st Qu.: -3.1250

Median :10.00 Median : 6.00 Median : -1.000 Median : 0.0000

Mean :10.79 Mean : 7.81 Mean : -2.181 Mean : -0.3275

3rd Qu.:16.00 3rd Qu.:12.00 3rd Qu.: 1.000 3rd Qu.: 2.9250

Max. :44.00 Max. :35.00 Max. : 20.000 Max. : 19.4000

Min Maj Match Misc

Min. : 0.000 Min. : 0.000 Min. :0.000000 Min. :0.0000

1st Qu.: 3.750 1st Qu.: 0.000 1st Qu.:0.000000 1st Qu.:0.0000

Median : 9.000 Median : 0.000 Median :0.000000 Median :0.0000

Mean : 9.995 Mean : 1.099 Mean :0.005495 Mean :0.1703

3rd Qu.:15.000 3rd Qu.: 1.000 3rd Qu.:0.000000 3rd Qu.:0.0000

Max. :39.000 Max. :14.000 Max. :1.000000 Max. :4.0000

Game CF CA FF

Min. :0.00000 Min. : 8.0 Min. : 6.0 Min. : 5.0

1st Qu.:0.00000 1st Qu.: 265.5 1st Qu.: 284.5 1st Qu.: 197.5

Median :0.00000 Median : 782.5 Median : 857.5 Median : 585.0

Mean :0.07143 Mean : 814.8 Mean : 802.2 Mean : 607.3

3rd Qu.:0.00000 3rd Qu.:1280.2 3rd Qu.:1221.8 3rd Qu.: 953.5

Max. :2.00000 Max. :2308.0 Max. :2009.0 Max. :1668.0

FA SF SA xGF

Min. : 5.0 Min. : 4.0 Min. : 2.0 Min. : 0.20

1st Qu.: 218.8 1st Qu.: 139.2 1st Qu.: 156.0 1st Qu.: 11.38

Median : 633.5 Median : 424.0 Median : 465.5 Median : 36.90

Mean : 596.7 Mean : 437.6 Mean : 428.6 Mean : 39.12

3rd Qu.: 917.2 3rd Qu.: 690.5 3rd Qu.: 664.8 3rd Qu.: 61.92

Max. :1510.0 Max. :1181.0 Max. :1073.0 Max. :111.10

xGA SCF SCA GF

Min. : 0.40 Min. : 0.0 Min. : 0.0 Min. : 0.00

1st Qu.:12.75 1st Qu.: 36.0 1st Qu.: 41.0 1st Qu.: 9.00

Median :39.40 Median :122.0 Median :130.0 Median : 34.00

Mean :38.07 Mean :129.9 Mean :126.7 Mean : 38.22

3rd Qu.:57.85 3rd Qu.:204.5 3rd Qu.:197.0 3rd Qu.: 64.00

Max. :97.20 Max. :419.0 Max. :344.0 Max. :120.00

GA RBF RBA RSF

Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 0.00

1st Qu.: 11.00 1st Qu.: 9.00 1st Qu.:10.00 1st Qu.: 12.00

Median : 38.00 Median : 28.00 Median :28.00 Median : 37.00

Mean : 36.48 Mean : 31.35 Mean :29.52 Mean : 38.38

3rd Qu.: 56.00 3rd Qu.: 48.00 3rd Qu.:44.00 3rd Qu.: 59.00

Max. :100.00 Max. :110.00 Max. :95.00 Max. :130.00

RSA DSF DSA FOW

Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. : 4.0

1st Qu.: 13.00 1st Qu.: 22.00 1st Qu.: 25.00 1st Qu.: 135.8

Median : 39.00 Median : 67.00 Median : 71.00 Median : 438.0

Mean : 38.07 Mean : 69.73 Mean : 67.59 Mean : 432.0

3rd Qu.: 57.00 3rd Qu.:107.00 3rd Qu.:101.25 3rd Qu.: 675.2

Max. :112.00 Max. :213.00 Max. :185.00 Max. :1257.0

FOL HF HA GVA

Min. : 4.0 Min. : 0.0 Min. : 2.0 Min. : 0.00

1st Qu.: 149.0 1st Qu.:146.5 1st Qu.:123.5 1st Qu.: 41.75

Median : 451.0 Median :348.0 Median :330.0 Median :124.00

Mean : 428.7 Mean :325.1 Mean :314.5 Mean :128.77

3rd Qu.: 664.8 3rd Qu.:474.0 3rd Qu.:465.0 3rd Qu.:200.25

Max. :1196.0 Max. :926.0 Max. :870.0 Max. :388.00

TKA PENT PEND OPS

Min. : 0.00 Min. : 0.00 Min. : 0.00 Min. :-1.500

1st Qu.: 30.00 1st Qu.: 18.75 1st Qu.: 19.00 1st Qu.: -0.100

Median : 88.50 Median : 54.00 Median : 52.00 Median : 0.400

Mean : 96.08 Mean : 50.26 Mean : 49.34 Mean : 1.316

3rd Qu.:147.00 3rd Qu.: 75.25 3rd Qu.: 75.00 3rd Qu.: 2.225

Max. :347.00 Max. :122.00 Max. :127.00 Max. :10.500

DPS PS OTOI Grit

Min. :-0.200 Min. :-1.200 Min. : 33.51 Min. : 0.00

1st Qu.: 0.200 1st Qu.: 0.200 1st Qu.: 948.12 1st Qu.: 55.75

Median : 1.050 Median : 1.800 Median :2560.87 Median :130.50

Mean : 1.383 Mean : 2.704 Mean :2088.29 Mean :141.52

3rd Qu.: 2.000 3rd Qu.: 4.600 3rd Qu.:3056.93 3rd Qu.:208.00

Max. : 7.200 Max. :12.600 Max. :3521.78 Max. :622.00

DAP Pace GS GS/G

Min. : 0.000 Min. : 77.6 Min. : -3.500 Min. : -0.8100
1st Qu.: 5.000 1st Qu.: 104.7 1st Qu.: 3.075 1st Qu.: 0.1400
Median : 7.650 Median : 109.0 Median : 17.000 Median : 0.2950
Mean : 9.398 Mean : 109.0 Mean : 22.379 Mean : 0.3304
3rd Qu.: 12.000 3rd Qu.: 114.2 3rd Qu.: 37.100 3rd Qu.: 0.5000
Max. : 52.500 Max. : 175.7 Max. : 99.200 Max. : 1.2600